DSA211 Group Project Proposal (Revised)

Group G1-4 | Daniel, Nicholas, Adam, Aarav, Imran | Topic 3: Model Complexity and the Bias–Variance Trade-Off (Monte Carlo Study)
Revised after the instructor's feedback (“Approved with changes”).

1. Research question

For polynomial regression fitted by OLS with degree $d = 1, \ldots, 10$, how do bias², variance and irreducible error contribute to expected test error? Specifically:

1. How do bias² and variance change with d, and does bias² + variance + σ² reproduce the expected test MSE?
2. Which degree minimises expected test error, and does it match the true degree of 3?
3. How does the optimal degree change as the noise level σ increases (σ = 0.2, 0.3, 0.5, 1), and what does this imply for real data where σ is unknown?
4. When polynomial regression is replaced by a LASSO path, with effective degrees of freedom as the complexity axis, how does the bias–variance profile compare with polynomial OLS?

Course methods applied (the brief requires at least three): polynomial regression by OLS (Weeks 2–4); the bias–variance decomposition of expected test error (Week 1); train/test discipline and k-fold cross-validation for choosing the polynomial degree, with the one-standard-error rule (Weeks 7–8); the LASSO with standardised predictors and λ chosen by cross-validation (Week 9). Topic 3 is a simulation study, so the data are generated from the DGP below instead of downloaded.

2. Data-generating process

- True function $f(x) = x^3 - x$ (true degree 3). The “$-x$” term gives a visible S-shape, so degrees 1–2 are clearly too simple and degrees 4–10 are more flexible than needed.
- $X \sim U(-1, 1)$. A domain starting at 0 would make the cubic look quadratic.
- $\varepsilon \sim N(0, \sigma^2)$, independent of X; $Y = f(X) + \varepsilon$. Baseline σ = 0.3; noise extension σ = 0.2, 0.5, 1.
- Training size $n = 50$ per run; $R = 500$ simulation runs (the brief requires at least 200); degree range **1 to 10** everywhere (degree 0 is dropped).
- Extensions: **noise level** (Section 7) and **regularised family**, a LASSO path on the powers $x, \ldots, x^{10}$ (Section 8). Both are carried out in depth with predictions stated in advance.

3. Signal-to-noise ratio (stated before running)

On $U(-1, 1)$, $E[f(X)] = 0$ and $\mathrm{Var}(f(X)) = 1/7 - 2/5 + 1/3 = 8/105 \approx 0.076$, so $\mathrm{SD}(f) \approx 0.276$. Define $\mathrm{SNR} = \mathrm{Var}(f)/\sigma^2$.

σ	0.2	0.3 (baseline)	0.5	1.0
σ²	0.04	0.09	0.25	1.00
SNR	1.90	0.85	0.30	0.076

The baseline σ = 0.3 already equals the signal SD. At σ = 1 the signal is nearly invisible, so a simpler model may win. We treat that as an expected result, not a failure of the simulation.

4. Predictions stated before running

For OLS with $p = d + 1$ coefficients, the prediction variance at x_0 (given the design X) is $\sigma^2 \cdot v(x_0)^\top (X^\top X)^{-1} v(x_0)$, where $v(x_0)$ is the polynomial basis evaluated at x_0; averaged over x_0 it is approximately $\sigma^2 p/n$. The best straight-line approximation of f is $-(2/5)x$, leaving $x^3 - 0.6x$

unexplained, and adding x^2 does not help because f is an odd function. So the population bias² of the best degree-1 or degree-2 fit is 4/175 ≈ 0.023, and for $d \geq 3$ it is 0.

ID	Prediction
H1	**Bias².** About 0.023 for $d = 1$ and 2, then about 0 for $d \geq 3$. It does **not** depend on σ.
H2	**Variance.** Increases with d. For $d \geq 3$, where OLS is unbiased for f, variance scales **exactly** with σ² (Var/σ² is identical at every noise level). For $d < 3$ the misspecified fit also varies with the random design, adding a small σ-free component.
H3	**Irreducible error** equals σ² at every degree.
H4	**Optimal degree.** Approximate excess error is $0.023 + (d+1)\sigma^2/n$ for $d \in \{1, 2\}$ and $4\sigma^2/n$ for $d = 3$. Degree 3 beats degree 1 when $\sigma < \approx 0.76$. Predicted optimum: **degree 3** for σ = 0.2, 0.3, 0.5 and **degree 1** for σ = 1.
H5	**Edges.** Variance is smallest near $x_0 = 0$ and rises toward ±1 with leverage, more steeply for higher d. This is a leverage effect, not exponential growth.
LASSO extension. Effective degrees of freedom (df) are estimated by the number of nonzero coefficients plus 1 for the intercept; Zou, Hastie & Tibshirani (2007) show the nonzero count is an unbiased estimate of the LASSO's degrees of freedom. For polynomial OLS, df = $d + 1$.	
H6	**Profile.** As λ falls, df rises continuously; bias² falls and variance rises, giving the same U-shaped test MSE as the polynomial family but on a continuous complexity axis.
H7	**Cost of shrinkage at low noise.** The true function uses only x and x^3, but the LASSO also shrinks those coefficients, and the standardised powers are strongly correlated (on U(−1, 1), $\text{corr}(x, x^3) \approx 0.92$). So at low noise the LASSO's best test MSE is expected to be **no better** than degree-3 OLS, which is unbiased with only 4 coefficients.
H8	**Noise.** As σ grows, the best λ grows (lower df) and the LASSO's position relative to polynomial OLS improves, because shrinkage buys a large variance cut for a little bias. At σ = 1 the LASSO may beat the best polynomial.
H9	**df estimate.** Nonzero count + 1 agrees with the covariance definition, $\text{df} = \Sigma_i \text{Cov}(\hat{y}_i, y_i)/\sigma^2$.

5. Method

1. **Fix the test inputs once.** Draw 200 test inputs $x_{0,1..200}$ from U(−1, 1) a single time. No noise is attached to them at this stage.
2. **Simulation runs.** In each of $R = 500$ runs, draw 50 training points, fit polynomial OLS of degree 1–10 and predict at the fixed x_0. Fits use the course pipeline (standardise x, then take its powers, then OLS; Lab 7), which keeps high-degree fits accurate. The same X and standardised noise are reused at every σ and the models are refitted at each σ, so differences between σ values are not Monte Carlo luck.
3. **Bias² and variance are computed against $f(x_0)$, not a noisy y.** With $\hat{y}_r(x_0)$ the run-r prediction and $\bar{y}$ their mean over runs:
   $\text{Var}(x_0) = (1/(R-1))\ \Sigma_r\ (\hat{y}_r - \bar{y})^2$; $\text{Bias}^2(x_0) = (\bar{y} - f(x_0))^2 - \text{Var}(x_0)/R$.
   The second term removes the upward bias of the plug-in estimator, whose expectation is Bias² + Var/R. Both are then averaged over the 200 test inputs, and their Monte Carlo standard errors are estimated by bootstrapping the 500 runs. Irreducible error is σ², which we choose, so it is not estimated.
4. **Expected test MSE with fresh noise in every run.** Draw new ε^*_r at x_0 in every run: $y^*_{rt} = f(x_{0,t}) + \varepsilon^*_{rt}$. Then
   $\text{MSE}(d) = (1/(R \cdot 200))\ \Sigma_r\ \Sigma_t\ (\ y^*_{rt} - \hat{y}_{rt}(d)\ )^2$ (squared errors).
   We verify Bias² + Variance + σ² ≈ MSE(d) for every degree and report the Monte Carlo standard error of MSE(d).

5. **Visualise and locate the optimum.** Plot Bias², Variance, σ² and test MSE against degree, mark the degree with the lowest MSE, and compare it with the true degree 3. Because every degree is fitted on the same training sets, we judge whether neighbouring degrees are distinguishable from the **paired** run-by-run difference in test MSE and its standard error.
6. **Variance against x_0.** Plot Var(x_0) against x_0 for several degrees (e.g. 1, 3, 6, 10) and overlay σ² × the average leverage.
7. **Noise extension.** Repeat for σ = 0.2, 0.5, 1 and test H1, H2 and H4 (see Section 7).

6. Modelling workflow (brief's requirement)

The Monte Carlo study is an oracle analysis: it reports true expected error and makes no data-driven selection. Wherever we actually choose a model from data (the family, the degree or λ), we follow the brief's workflow on a simulated dataset (N = 100, σ = 0.3):

1. Split once, at the start (80% train / 20% test).
2. Compare all candidates only by cross-validation inside the training set, on the same folds: polynomial OLS of degree 1–10, and the LASSO on the standardised powers x, …, x^{10} over the 40 penalties of Section 8 (standardisation learned inside each training fold).
3. **CV metric, stated before the result:** 10-fold cross-validated mean squared prediction error. Primary rule: the single candidate with the lowest CV MSE across both families; the one-standard-error choice within each family is also reported, and the group states which model it adopts before the test set is used.
4. Touch the test set exactly once, on the single chosen model.
5. Refit the chosen specification on all N observations for the final coefficients.

We then compare the CV choice with the oracle optimum from a Monte Carlo run at the matching training size (n = 80). We also run the selection procedure alone (5-fold CV, no test set) on 200 independent datasets of n = 50 at each σ, to see how often each rule recovers the oracle degree.

7. Noise-level extension and testable predictions

- **Bias² is σ-free:** the bias² curves for σ = 0.2, 0.3, 0.5, 1 should coincide (up to Monte Carlo error, which is of order Var/R).
- **Variance scales with σ²:** Var/σ² should be the same at every σ for $d \geq 3$ (exactly, because the same draws are reused at every σ). For $d < 3$, Var/σ² falls as σ grows because of the σ-free design component. We also compare the empirical variance with σ² × average leverage.
- **Optimal degree:** predicted 3 for σ ≤ 0.5 and 1 for σ = 1 (H4), with the crossover near σ ≈ 0.76. We will report the empirical optimum for each σ next to the prediction.
- **Implication for real data:** if σ is large relative to the signal, even the true model can be too complex to estimate well from limited data, and σ is not known in practice. This is why selection has to be by cross-validation (Section 6).

8. LASSO-path extension (regularised family)

1. **Same simulation design.** In each of the same 500 runs, the LASSO receives the 10 powers x, …, x^{10} of the training inputs, standardised with that run's training mean and SD (the penalty is not scale-invariant). As in Lab 9, the LASSO is fitted separately at each of 40 penalties, with an unpenalised intercept, and every fit predicts at the same fixed x_0 and is scored against the same fresh test noise as polynomial OLS, at every σ.
2. **Decomposition.** Bias², variance, σ² and test MSE are computed for every penalty exactly as in Section 5, steps 3–4, with the same identity check and Monte Carlo standard errors.
3. **Complexity axis.** df(λ) = average over runs of (nonzero count + 1). Both families are plotted on the same df axis, so their bias–variance profiles can be compared directly (H6).

4. **Checking the df estimate (H9).** For one fixed set of training inputs, 400 fresh noise vectors are drawn, the LASSO is refitted at every penalty each time, and $\Sigma_i \mathrm{Cov}(\hat{y}_i, y_i)/\sigma^2$ is estimated and compared with the nonzero count + 1. As a check of the estimator, degree-3 OLS should give df ≈ 4.
5. **Best LASSO vs best polynomial (H7, H8).** At each σ we compare the lowest test MSE of each family. Because both use the same runs and test noise, the difference is computed run by run, with a paired standard error.
6. **Which powers are kept.** At each σ's best penalty we report how often the LASSO keeps x, x^3, exactly $\{x, x^3\}$, and the most common sets of powers, to explain its bias–variance profile.

9. Potential issues and remedies

Issue	Remedy
Plug-in Bias² is inflated by Var/R when R is small.	Bias-corrected estimator (Section 5, step 3); R = 500; Monte Carlo standard errors for bias², variance (bootstrap over runs) and test MSE.
Test MSE from one fixed noise draw is itself noisy.	Fixed x_0, but fresh noise at x_0 in every run; MSE averaged over runs.
Variance is larger near ±1: an edge point has training data on one side only, so its leverage is high.	Not removed; it is reported. Shown by plotting variance against x_0 with σ² × average leverage overlaid.
Numerical conditioning of degree 10 fits with n = 50. (On [−1, 1], x^{10} is at most 1, so values do not blow up; the risk is near-collinear columns.)	Standardise x before taking powers (Lab 7 pipeline); checked against numpy's ordinary polynomial fit for every degree on 20 runs.
Very heavy-tailed variance at degrees 8–10 makes their MSE estimates noisy.	Report standard errors; use paired comparisons between degrees; clip plot axes for readability; do not over-interpret differences among degrees 8–10.
At σ = 1 a low degree may win.	Predicted in advance (H4); reported as a result.
The LASSO penalty depends on the scale of each power.	Standardise the powers with the training mean and SD in every run, and inside every CV fold (Week 9 practice).
Coordinate descent may not converge at very small penalties, where the correlated powers are barely penalised.	High iteration limit and tight tolerance; the number of non-converged fits is reported, and the penalties where they occur are checked to lie well below every σ's best penalty.
Strongly correlated powers make the LASSO's chosen set of powers unstable from sample to sample.	Report how often each power and each set is chosen, rather than a single selected model.
The nonzero count is only an estimate of the LASSO's degrees of freedom.	Checked against the covariance definition of df (Section 8, step 4).

10. Response to the instructor's feedback

#	Comment	Change made
1	Separate fixed test inputs from the noise.	Fix x_0 once; bias² and variance computed against $f(x_0)$; fresh noise at x_0 in every run for test MSE (Section 5).
2	Check the Step 4 formula.	The old formula summed signed errors without squaring. Replaced by the mean of squared errors, MSE(d) in Section 5, step 4.

3	Degree range consistency.	Degrees 1–10 throughout; degree 0 removed.
4	Report the signal-to-noise ratio.	Var(f) = 8/105, SD(f) ≈ 0.276, SNR table, and the expected low-degree winner at σ = 1 stated before running (Sections 3–4).
5	Use $\sigma^2 \cdot x_0^\top (X^\top X)^{-1} x_0$ as a testable prediction.	Variance ∝ σ² and σ-free bias² become H1–H2 and tests in Section 7. We note that exact σ² scaling holds for a fixed design, and in our random design for $d \geq 3$; for $d < 3$ a small σ-free component remains.
6	Variance near ±1 is leverage, not exponential.	Reworded (H5); variance vs x_0 plot for several degrees with the leverage overlay (Section 5, step 6).
—	Follow the brief's workflow; add the AI declaration.	Section 6 and the declaration below.

11. Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used **Claude (Anthropic)** for **writing and debugging Python code for the simulation notebook and restructuring this proposal in response to instructor feedback**. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published work.
(Draft wording for the group to confirm or correct.)